

Handout 1

Evaluation Criteria

Software Verification

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Version of October 5, 2026

1 Summary of Exercise Grades

Exercise 1	definition of <code>red_expr</code>	1 point
Exercise 2	<code>red_expr_star</code>	1 point
Exercise 3	<code>red_asm</code>	2 points
Exercise 4	<code>red_asm_star</code>	2 points
Exercise 5	<code>compile</code>	2 points
Exercise 6	<code>red_expr_star_const_inv</code>	2 points
Exercise 7	<code>red_asm_compile_op</code>	2 points
Exercise 8	<code>comm_red_asm_append</code>	3 points
Exercise 9	<code>compile_correct_gen</code>	3 points
Exercise 10	<code>compile_correct</code>	2 points
	TOTAL	20 points

Table 1: Exercises Grades.

Table 1 summarizes the grades for each exercise in Handout 1. In lemma `compile_correct_gen` you only full points if you *avoid doing case analysis on the binary operator $\circ$* , when an expression is of the form $e_1 \circ e_2$.

2 Proof Organization

- If a proof is divided in n cases, each case counts $1/n$ points of the exercise grade.
- If a case as m sub-cases, each sub-case counts $1/m$ points of the case grade.
- If a proof is not clearly organized in cases and sub-cases, it loses 20% of the exercise total grade.

- A proof that unnecessarily introduces superfluous cases (*e.g.*, doing **induction** when the proof does not require induction) loses 20% of the exercise total grade.

3 Proof Readability

You should indent and comment your proof, indicating the main cases and sub-cases. A proof that is hard to read loses 5% of the exercise total grade.

4 Proof Automation and Conciseness

You should strive for as much automation as possible in our proofs. This means using tactics such **lia**, **auto**, or **assumption** as soon as possible. If a case within a proof is longer than required, it loses 10% of that specific case grade.